

Department of Computer Science and Engineering

III/II CSE - R23 Regulations

BIG DATA ANALYTICS

Question Bank with Blooms Taxonomy Levels

Unit I: Introduction to Big Data & Hadoop

- 1) **Analyze** the convergence of key trends that have led to the explosion of Big Data in the modern era. (Level 4)
- 2) **Explain** the role of Big Data analytics in algorithmic trading and credit risk management. (Level 2)
- 3) **Illustrate** how Big Data technologies are revolutionizing the healthcare industry with specific examples from medicine. (Level 3)
- 4) **Compare and contrast** structured and unstructured data, providing industry-specific examples for each. (Level 4)
- 5) **Discuss** the impact of Big Data on marketing and web analytics for modern businesses. (Level 2)
- 6) **Analyze** the importance of fraud detection and risk management using Big Data frameworks. (Level 4)
- 7) **Explain** the concept of "inter and trans firewall analytics" in the context of business intelligence. (Level 2)
- 8) **Evaluate** the role of Crowdsourcing analytics and mobile business intelligence in current data trends. (Level 4)
- 9) **Apply** the fundamental principles of open-source technologies to explain the growth of the Hadoop ecosystem. (Level 3)
- 10) **Explain** the relationship between Cloud computing and Big Data processing capabilities. (Level 2)

Unit II: NoSQL & Cassandra

- 11) **Analyze** the limitations of traditional RDBMS and explain why NoSQL aggregate data models are preferred for Big Data. (Level 4)
- 12) **Compare** Key-Value, Document, and Graph data models, citing appropriate use cases for each. (Level 4)
- 13) **Explain** the concepts of sharding and distribution models in NoSQL databases. (Level 2)
- 14) **Analyze** the difference between Master-Slave replication and Peer-to-Peer replication in distributed systems. (Level 4)
- 15) **Discuss** the CAP theorem and the necessity of relaxing consistency in large-scale NoSQL deployments. (Level 2)
- 16) **Explain** the significance of version stamps and materialized views in maintaining data integrity. (Level 2)
- 17) **Apply** Cassandra Query Language (CQL) concepts to demonstrate table creation and data loading procedures. (Level 3)
- 18) **Analyze** the schema-less nature of NoSQL databases and its impact on application development. (Level 4)
- 19) **Explain** the internal architecture of Cassandra and how it handles high availability. (Level 2)
- 20) **Illustrate** the process of reading and writing data in a distributed Cassandra cluster. (Level 3)

Unit III: Hadoop HDFS, MapReduce & Hive

- 21) **Analyze** the architecture of the Hadoop Distributed File System (HDFS), focusing on NameNodes and DataNodes. (Level 4)
- 22) **Explain** how Hadoop achieves fault tolerance and high availability through data replication. (Level 2)
- 23) **Discuss** the principle of Data Locality and its importance in reducing network congestion. (Level 2)
- 24) **Analyze** the MapReduce architecture and describe the end-to-end data flow during a job execution. (Level 4)
- 25) **Apply** Hadoop I/O concepts to explain data integrity, compression, and serialization techniques. (Level 3)
- 26) **Compare** different Hive data types and file formats used for large-scale data warehousing. (Level 4)
- 27) **Analyze** the various Join strategies available in Hive and their performance implications. (Level 4)
- 28) **Explain** the benefits of Table Partitioning and Bucketing in Hive for query optimization. (Level 2)
- 29) **Apply** HiveQL Data Manipulation (DML) and Definition (DDL) commands to manage structured data. (Level 3)
- 30) **Analyze** the role of Window functions in Hive for performing complex analytical queries. (Level 4)

Unit IV: Apache Spark Core & Deployment

- 31) **Analyze** the fundamental advantages of Apache Spark over Hadoop MapReduce, specifically regarding iterative workloads. (Level 4)
- 32) **Explain** the concept of Lazy Evaluation and how it allows Spark to optimize the execution plan. (Level 2)
- 33) **Analyze** the Directed Acyclic Graph (DAG) and distinguish between Jobs, Stages, and Tasks. (Level 4)
- 34) **Compare** the roles of SparkContext and SparkSession in modern Spark applications. (Level 4)
- 35) **Differentiate** between Narrow and Wide transformations, providing examples of each and their impact on shuffles. (Level 4)
- 36) **Apply** the Catalyst Optimizer's four phases (Analysis, Logical Optimization, Physical Planning, and Code Gen) to a sample SQL query. (Level 3)
- 37) **Explain** the use of Broadcast Variables and Accumulators in distributed shared memory processing. (Level 2)
- 38) **Analyze** the differences between deploying Spark on Standalone mode versus Spark on YARN. (Level 4)
- 39) **Illustrate** the process of converting an RDD to a DataFrame and explain why DataFrames are generally faster. (Level 3)
- 40) **Explain** how to use the Spark UI and History Server for debugging and identifying performance bottlenecks like data skew. (Level 2)

Unit V: Spark Performance & Structured Streaming

- 41) **Analyze** various Spark performance tuning methods, including memory management and partition sizing. (Level 4)
- 42) **Explain** the fundamentals of Stream Processing and the difference between Event-Time and Processing-Time. (Level 2)
- 43) **Analyze** the necessity of Stateful Processing in real-time analytics. (Level 4)
- 44) **Compare** Tumbling Windows, Sliding Windows, and Session Windows in the context of Event-Time processing. (Level 4)
- 45) **Apply** the concept of Watermarks to handle late-arriving data in a streaming pipeline. (Level 3)
- 46) **Explain** the core concepts of Structured Streaming and its "continuous processing" model. (Level 2)
- 47) **Analyze** the mechanism for dropping duplicates in a live data stream to ensure data quality. (Level 4)
- 48) **Apply** various transformations on streams and describe the available Input and Output sinks. (Level 3)
- 49) **Discuss** the implementation of Structured Streaming in action for a real-world use case. (Level 2)
- 50) **Analyze** the impact of windowing operations on resource consumption and latency in Spark Streaming. (Level 4)
